AUDIT SURFACE FOR

Capability Matters: A Casebook

Validation & Audit

INDEXES · PER-CASE REFERENCES

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SECTION

What this document is for

Validation and Audit is the auditor-facing companion to **Capability Matters: A Casebook**. It carries the surfaces an instructor, reviewer, or accreditation reader needs to verify the casebook — the indexes that let any case be located by domain or by LENS course, and the per-case reference appendix that lets any source be traced. It is the same data the casebook itself ships, lifted into a standalone document so the LENS Companion can stay focused on what LENS **is** (the five competencies, the CLOs, the course mapping) without the audit pages travelling with it.

The document is in three parts:

I. Cases by primary domain — every case classified by the operational domain it primarily evidences, so a domain expert can find the cases that touch their territory without scanning the table of contents.

II. Cases by LENS course — every case classified by the LENS course it serves as a worked example for, so syllabus authors and capstone advisors can find the cases that fit a course's stated learning outcomes.

III. References, case by case — every reference cited in every case, organised by case in number order, with an explicit **Retrieved from:** line per entry so a reviewer can locate the canonical source.

> **On verification.** The running per-case verification track lives in the repository at `casebook/verification-log.md`, a 194-row table whose six auto-prefilled columns track the structural and sourcing checks the build pipeline can answer on its own. The seventh column — **conclusions reasonable** — is the human case-by-case content read this document supports. The references appendix below is the per-case checklist for that read; the **Retrieved from:** lines are the canonical-access record each case acquires as the verification pass closes.

DOMAIN INDEX

The cases by domain.

The matrix at the front lists every case in number order. This index reorganises the cases by primary domain — a reader following healthcare, defense, education, or any other thread can find the relevant cases together. Each case appears once, under its primary domain. Where a section opens with a callout, it names the dataset's standout failure and, where one exists in that domain, its standout success.

DEFENSE

STANDOUT FAILURE	Case 5	**Operation Eagle Claw (1980).** Cross-organizational capability that existed on no service's deliverable list; the mission was improvised across the Army, Navy, Marines, Air Force, and CIA. Produced the Holloway Commission, USSOCOM, and Goldwater-Nichols.
STANDOUT SUCCESS	Case 18	**U.S. Nuclear Navy / Rickover Training Model (1954–).** The longest continuous capability-engineering program in any high-consequence domain. Sixty-plus years of zero reactor accidents through training treated as a system parameter.

124	USS Fitzgerald & USS John S. McCain	2017	T·K·N
126	F-35 Sustainment & Maintainer Shortage	ongoing	T·K·D
127	Military Fratricide — Desert Storm to Afghanistan	1991 – 2014	T·H·K
128	Operation Eagle Claw	1980	T·K
129	Patriot Missile / Dhahran	1991	D·H·K
130	USS Vincennes & Iran Air Flight 655	1988	H·T
131	Stanislav Petrov / 1983 False Alert	1983	H·T
133	Mark 14 Torpedo Failures	1941 – 1943	T·K·G
134	V-22 Osprey	1991 – present	T·H·N
135	9/11 Intelligence Sharing Failures	1996 – 2001	G·K
137	U.S. Nuclear Navy / Rickover Training Model	1954 – present	T·K·N
138	MIL-STD-1472H — Human Engineering as a Binding Acquisition Standard	2020 revision (series since 1968)	G·K

AVIATION

STANDOUT FAILURE — **Case 7**

Boeing 737 MAX / MCAS (2018–19). A single-string angle-of- attack sensor, a flight-control law that assumed pilot mental models from prior models, and a certification chain that did not knit the pieces together. 346 lives.

STANDOUT SUCCESS — **Case 70**

Crew Resource Management & CAST (1981–). First-officer authority engineered into the cockpit, a continuous learning architecture across the industry, and the safety record that followed.

HEALTHCARE

STANDOUT FAILURE	Case 61	**EHR / CPOE Implementation (2005–).** Roughly $30B federal investment in systems designed for billing and administration, deployed against clinical workflow. Usability is now itself a patient-safety variable at scale.
STANDOUT SUCCESS	Case 109	**WHO Surgical Safety Checklist (2008–).** One page, nineteen items, paired with a required pause. Halved surgical mortality in the multi-site pilot; the smallest effective capability intervention in the dataset.

1	Therac-25 1985 – 1987	H·D·G
2	EHR / CPOE Implementation 2005 – present	H·D·G
4	Mid Staffordshire NHS Foundation Trust 2005 – 2009	G·N·K
5	Epic Sepsis Model 2017 – 2021	D·G·N
8	Medical Errors as Systemic Failure 1999 – present	T·H·N·K·G
9	Vioxx Withdrawal 1999 – 2004	G·D
11	California Nurse Staffing Ratios — Legislating a Capability Requirement 2004 – 2010	G·K
12	ACGME 80-Hour Resident Duty-Hour Reform 2003–2017	T·K·N
13	Implementation Science in Healthcare — The 17-Year Gap ongoing	K·G·N
19	Keystone ICU / Pronovost Checklist 2004 – present	T·N
20	TREWS / Sepsis Watch 2018 – 2022	H·K·G
21	Sterile-Cockpit Ward Rounds — Adapting an Aviation Principle to Clinical Handoff 2024 – 2025	H·K·N
22	ChatGPT in Healthcare — Hallucination Cases 2023 – present	H·D
23	WHO Surgical Safety Checklist 2008 – present	T·N
24	Bristol Heart Babies Reform 1984 – present	G·N
32	Annual-Screening UI Redesign + CDS at University of Missouri Health Care 2020	T·D·N
33	Alert-Fatigue Redesign — Cutting EHR Alerts Without Cutting the Safety Signal 2019 – 2024	T·D·N
34	COMPOSER Sepsis Prediction — The Third Clinical-AI Sepsis Case 2022 – 2024	T·K·D
35	Radiology AI Miscalibration 2018 – present	H·K·D
36	AlphaFold — Protein Structure Prediction 2020 – present	T
37	DeepMind Mammography — High-Profile Nature Paper, Replicability Pushback 2020	T·K·D

39	TeamSTEPPS 2006 – present	T·N

EDUCATION AT SCALE

STANDOUT FAILURE	Case 149	**Summit Learning / Personalized Learning Rollout (2014–19).** A defensible pedagogical platform, well funded, deployed across 380 schools — and pulled by parent revolt because the governance architecture (consent, evidence, exit) was never engineered alongside it.
STANDOUT SUCCESS	Case 110	**Georgia State University Predictive Analytics (2012–).** 800 risk factors per student, advising triggered by early warning signs, equity as a primary constraint. Six-year graduation 32% → 54%; equity gaps eliminated.

45	Tennessee Voluntary Pre-K Study 2009–2018	G·D
46	Algorithmic Bias in Educational Predictive Analytics ongoing	G·H·D
51	Atlanta Public Schools Cheating Scandal 2009 – 2015	G·N
53	inBloom 2014	G
54	Summit Learning / Personalized Learning Rollout 2014–2019	G·T·K
67	Cognitive Tutor / Carnegie Learning 1990s – present	T
80	Georgia State University Predictive Analytics 2012 – present	T·K
142	xAPI / Total Learning Architecture — Interoperability Gap ongoing	K·G
205	The Discipline We Build Next ongoing	T·K·N·G·H·D

ENERGY

STANDOUT FAILURE	Case 69	**Three Mile Island (1979).** Training matched the design-basis events the engineers had imagined; the accident was not one of them. The control-room interface confused command signal with valve state. Catalysed industry-wide reform.
STANDOUT SUCCESS	Case 172	**INPO and the Nuclear Academy (1979–).** The industry built the institution TMI revealed it needed. No INES-level event at U.S. commercial reactors since.

146	Northeast Blackout 2003	H·K

GOVERNMENT

STANDOUT FAILURE	Case 162	**Australia Robodebt (2016–20).** An income-averaging algorithm that reversed the burden of proof onto welfare recipients. The Royal Commission found "venality, incompetence and cowardice"; the scheme was linked to deaths, including suicide.
NOTE	Case 111	No paired-intervention success in the government dataset. The reforms that follow these cases — Royal Commissions, statutory restructures — are documented in the narratives but did not produce a LENS-style success case in time for this volume.

7	VA Wait-Time Scandal	2014	G·K·N

INDUSTRIAL

STANDOUT FAILURE	Case 147	**Bhopal (1984).** Training, knowledge, normalisation of deviance, and governance all collapsed in the same plant. Roughly 4,000 immediate deaths; chronic harms continue. The cautionary case for outsourced operational responsibility.
STANDOUT SUCCESS	Case 73	**Toyota Production System / Andon Cord (1950s–).** Any operator can stop the line. The smallest authority-architecture intervention in the dataset, and the most durably copied.

144	Kodak Digital Camera Stagnation	1975–2012	D·K·N
147	Takata Airbag Inflators	2008 – 2023	D·G
149	Volkswagen Dieselgate	2015	D·G
151	GM Ignition Switch	2002 – 2014	D·G
155	Toyota Production System / Andon Cord	1950s – present	N·G

TECHNOLOGY

STANDOUT FAILURE	Case 66	**UK Post Office Horizon Scandal (1999–2015).** Institutional refusal to believe sub-postmasters' reports of computer error, sustained over fifteen years. Hundreds of wrongful convictions; ruined lives and at least four suicides — the longest-running operator-feedback failure in the dataset.
NOTE	Case 132	The technology dataset's success story is filed under Healthcare: AlphaFold (Case 36), the contemporary worked example of computational capability-engineering done as a discipline.

143	Knight Capital Trading Loss	2012	D·K

AI IN EDUCATION

K-12 EDUCATION

K-12 MATHEMATICS

K-12 SCIENCE

AIR TRAFFIC MANAGEMENT

ANESTHESIOLOGY

AVIATION INFRASTRUCTURE

AVIATION SAFETY

CLINICAL CARE

CLINICAL MEDICINE

CLINICAL/TRANSLATIONAL RESEARCH

CORPORATE L&D

ED-TECH

EDUCATION (NORWAY)

EDUCATION AT SCALE

GENERAL AVIATION

GLOBAL HEALTH

GOVERNMENT

HEALTH PROFESSIONS EDUCATION

HEALTHCARE/ONCOLOGY

HIGHER EDUCATION

HIGHER EDUCATION (AFRICA)

HIGHER EDUCATION (BRAZIL)

HIGHER EDUCATION (LATIN AMERICA)

HIGHER ED (UK)

HIGHER-ED ANALYTICS

HOSPITAL PHARMACY

IMPLEMENTATION SCIENCE

LEARNING ANALYTICS

MEDICAL DEVICES

MEDICAL EDUCATION

MEDICAL-DEVICE REGULATION

MULTIMODAL LEARNING ANALYTICS

NEUROSCIENCE/CONNECTOMICS

NUCLEAR ENGINEERING

NUCLEAR POWER

154 INL Turbine-Control Upgrade — Low-Burden Cutover as a Human-Factors Finding 2014 G·K·H

NURSING EDUCATION

16 NCSBN National Simulation Study — Licensing the 50% Substitution Rule 2014 G·K·D

PHARMA SAFETY

38 iPLEDGE Isotretinoin REMS — When the Authorization Mechanism Underperforms 2006 – 2011 G·D·N

SOFTWARE ENGINEERING EDUCATION

71 Reflective Practice on a Work-Based Software Engineering Program — Longitudinal Capability Development 2025 (preprint) K·N

SPACE / AEROSPACE

STANDOUT FAILURE **Case 140** **Challenger & Columbia (1986 / 2003).** Normalisation of deviance across two decades and two crews, captured in the Rogers and CAIB reports as a culture that accepted flying with flaws when problems were defined as normal and routine.

NOTE **Case 18** No paired-intervention success in the aerospace dataset. See Defense (Rickover, Case 137) for the safety-engineering counterpart that aerospace did not build.

98 Mars Climate Orbiter — Unit Mismatch 1999 D·K

101 Ariane 5 Flight 501 1996 D·K·H

108 NASA EVA-23 — Water Intrusion Inside a Spacesuit 2013 H·N·D

111 Challenger & Columbia 1986 / 2003 N·K·G

113 Boeing Starliner 2019 – 2024 K·D

SURGERY

30 Surgical Skill Video Peer-Rating Predicts Complications 2013 H·D·K

31 Language of Surgery / JIGSAWS — Decomposing Skill into Measurable Units 2009 – 2016 T·K·H

TUTORING

77 Hybrid Human-AI Tutoring — Augmentation, Not Delegation 2024 T·K·D

WORKFORCE DEV

68 CIRCUIT — A Scalable, Equity-Centered Research Workforce Model 2017 – 2023 (six cycles) T·K

About this index. Each case is filed under its primary domain, so every case appears exactly once. Cases that span several domains (a healthcare-AI case is also a technology case, a defense-training case is also an education case) are cross-referenced in their narratives. The standout failures and successes are editorial picks — the case in that domain the editors return to most often in studio.

LENS COURSE INDEX

The cases by LENS course.

Each case names the LEN courses it serves as a worked example for. This index inverts that mapping: for every course in the specialization, the cases that inform it, in case-number order. A case appears under every course it supports, so many appear more than once.

LEN 1

25 cases

LEN 2

49 cases

LEN 3

37 cases

LEN 4

56 cases

LEN 5

66 cases

LEN 6

13 cases

LEN 7

79 cases

LEN 8

70 cases

LEN 9

18 cases

LEN 10

17 cases

APPENDIX · REFERENCES BY CASE

References, case by case.

Every reference cited in every case, organised by case in number order. The intent is that any reader can locate every source the book draws on. Each entry preserves the in-case citation form (author, year, venue) and adds an explicit **Retrieved from:** line pointing to the canonical access path — a publisher URL, a DOI, an agency landing page, or a stable archival locator. Where that line is left blank, the source is under verification; the running status of that work is in `casebook/verification-log.md`. Future editions will fill in remaining locators as the per-case review closes them out.

> **Style.** References are rendered in the form they appeared in the case — APA-style author/year/title/venue, with italicised book titles and journal names — supplemented by the explicit retrieval locator. Government and agency reports are cited by report number when one exists. Trade-book references that carry no DOI use ISBN as the stable identifier; that mapping is on the verification log.

CASE 1 Therac-25

1985 – 1987

1. N. G. Leveson & C. S. Turner, "An Investigation of the Therac-25 Accidents," IEEE Computer 26(7): 18–41 (1993). doi:10.1109/MC.1993.274940 — the removed hardware interlocks and the adapted software. — Retrieved from: ____________________

2. Leveson & Turner (1993) — the two software defects (data-entry race condition and counter overflow), the uninformative "Malfunction 54", overdoses up to 100×, six accidents, and at least three deaths. — Retrieved from: ____________________

3. Leveson & Turner (1993); N. G. Leveson, Safeware: System Safety and Computers (Addison-Wesley, 1995) — the manufacturer's denial and the systemic findings. — Retrieved from: ____________________

4. N. G. Leveson, Engineering a Safer World: Systems Thinking Applied to Safety (MIT Press, 2011) — why removing a safeguard requires explicitly reassigning its safety function. — Retrieved from: ____________________

5. The case's role in medical-device software regulation and software-safety practice (FDA software guidance; IEC 62304 lineage); see also Ethics Unwrapped, UT Austin. — Retrieved from: ____________________

CASE 2 EHR / CPOE Implementation

2005 – present

1. Health Information Technology for Economic and Clinical Health (HITECH) Act (2009) — the $30B EHR incentive program. — Retrieved from: ____________________

2. Y. Han et al., "Unexpected Increased Mortality After Implementation of a Commercially Sold CPOE System," Pediatrics 116(6): 1506–1512 (2005) — the disputed pediatric-ICU mortality result. — Retrieved from: ____________________

3. KLAS Arch Collaborative, EHR usability and safety surveys (2023) — usability as the top clinician complaint, correlated with safety outcomes. — Retrieved from: ____________________

4. D. Sittig & H. Singh, "Defining health information technology-related errors," and related EHR-safety work (2011–2013). — Retrieved from: ____________________

5. AHRQ Patient Safety Network, EHR-related harm cases — the default-result delay and the unactivated-alert overdose. — Retrieved from: ____________________

6. AMA / Pew / MedStar, EHR Usability and Patient Safety (2018); R. Wachter, The Digital Doctor (2015). — Retrieved from: ____________________

CASE 3 Watson for Oncology — Delegation Marketed Ahead of Capability 2013 – 2018

1. Ross & Swetlitz (2017–2018), "IBM Watson recommended unsafe and incorrect cancer treatments" and adjacent investigations, STAT News — the load-bearing primary source; investigative journalism drawing on leaked IBM internal documents. — Retrieved from: ____________________

2. Strickland (2019), "How IBM Watson Overpromised and Underdelivered on AI Health Care," IEEE Spectrum — independent retrospective synthesis of the public record. — Retrieved from: ____________________

3. Topol (2019), Deep Medicine, Basic Books — secondary academic situating of Watson within the broader clinical-AI delegation discourse. — Retrieved from: ____________________

4. v2 paired cases: TREWS (Case 20), Epic Sepsis Model (Case 5), SyRI (Case 189) — the AI-delegation typology. — Retrieved from: ____________________

CASE 4 Mid Staffordshire NHS Foundation Trust 2005 – 2009

1. R. Francis QC, Report of the Mid Staffordshire NHS Foundation Trust Public Inquiry (2013) — the staffing cuts and Foundation Trust targets. — Retrieved from: ____________________

2. Healthcare Commission, Investigation into Mid Staffordshire NHS Foundation Trust (2009) — ward conditions and excess mortality. — Retrieved from: ____________________

3. Francis QC (2013) — the 2,000-page report and 290 recommendations. — Retrieved from: ____________________

4. Francis QC (2013) — "the system as a whole failed in its most essential duty — to protect patients from unacceptable risks of harm" (quoted). — Retrieved from: ____________________

5. D. Berwick, A Promise to Learn — A Commitment to Act (National Advisory Group on the Safety of Patients in England, 2013). — Retrieved from: ____________________

6. K. Walshe & J. Higgins (2002) on NHS safety governance; cf. the VA wait-time scandal (Case 7). — Retrieved from: ____________________

CASE 5 Epic Sepsis Model 2017 – 2021

1. Wong et al. (2021), "External Validation of a Widely Implemented Proprietary Sepsis Prediction Model in Hospitalized Patients," JAMA Internal Medicine 181(8):1065–1070, doi:10.1001/jamainternmed.2021.2626. — Retrieved from: ____________________

2. Habib et al. (2021), commentary on Wong et al., JAMA Internal Medicine — on the implications for proprietary clinical AI. — Retrieved from: ____________________

3. FDA, Clinical Decision Support Software: Final Guidance (2022) — the post-Wong reframing of the EHR-embedded oversight question. — Retrieved from: ____________________

4. Adams et al. (2022), Nature Medicine — the paired positive case (20). — Retrieved from: ____________________

CASE 6 Racial Bias in Pain Assessment — The False-Belief Mechanism 2016

1. Hoffman, Trawalter, Axt, & Oliver (2016), "Racial bias in pain assessment and treatment recommendations, and false beliefs about biological differences between blacks and whites," PNAS 113(16):4296–4301, doi:10.1073/pnas.1516047113. — Retrieved from: ____________________

2. Anderson, Green, & Payne (2009), "Racial and ethnic disparities in pain: causes and consequences of unequal care," Journal of Pain 10(12):1187–1204 — the population-level disparity. — Retrieved from: ____________________

3. Sabin & Greenwald (2012), "The influence of implicit bias on treatment recommendations for 4 common pediatric conditions," American Journal of Public Health — the diffuse-mechanism backdrop the Hoffman precision improves on. — Retrieved from: ____________________

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CASE 7 VA Wait-Time Scandal 2014

1. VA Office of Inspector General, Report 14-02603-267 (2014) — secret waiting lists and falsified appointment data. — Retrieved from: ____________________

2. GAO Veterans Health Administration reports (2000–2019) — fifteen years of data-reliability warnings. — Retrieved from: ____________________

3. VA OIG reports (2005, 2007, 2008) — prior, unactioned findings. — Retrieved from: ____________________

4. D. Draper (GAO), House VA Committee testimony, GAO-19-687T (2019) — "schedulers are among the top ten highest-turnover positions in the VA" (quoted). — Retrieved from: ____________________

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6. C. Argyris & D. Schön, Organizational Learning (1978); measurement-gaming literature. — Retrieved from: ____________________

CASE 8 Medical Errors as Systemic Failure 1999 – present

1. Institute of Medicine, To Err Is Human: Building a Safer Health System (1999); Crossing the Quality Chasm (2001); Improving Diagnosis in Health Care (2015) — the field-defining trilogy and the 44,000–98,000 estimate; the systems framing. — Retrieved from: ____________________

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5. Bates, D. W., Levine, D. M., Salmasian, H., et al. (2023), "The Safety of Inpatient Health Care," NEJM 388(2):142–153 — eleven-hospital Massachusetts cohort; adverse events in 25% of admissions, 25% of those preventable; persistence of harm at scale. — Retrieved from: ____________________

6. Agency for Healthcare Research and Quality, National Healthcare Quality and Disparities Reports (annual); CDC WONDER ICD-coded mortality data — institutional context for the missing national active-surveillance instrument. — Retrieved from: ____________________

CASE 9 Vioxx Withdrawal 1999 – 2004

1. Bombardier, C. et al. (2000), "Comparison of upper gastrointestinal toxicity of rofecoxib and naproxen in patients with rheumatoid arthritis," VIGOR trial, NEJM 343(21):1520–1528 — the five-fold myocardial-infarction signal and the naproxen-protective hypothesis. — Retrieved from: ____________________

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CASE 11 California Nurse Staffing Ratios — Legislating a Capability Requirement 2004 – 2010

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4. Spetz, Chapman, Herrera, Kaiser, Seago, & Dower (2009), “Assessing the impact of California’s nurse staffing ratios on hospitals and patient care,” California HealthCare Foundation — implementation-period analysis. — Retrieved from: ____________________

CASE 12 ACGME 80-Hour Resident Duty-Hour Reform 2003–2017

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CASE 13 Implementation Science in Healthcare — The 17-Year Gap

ongoing

1. E. A. Balas & S. A. Boren (2000), Yearbook of Medical Informatics — the 17-year / 14% translation figures. — Retrieved from: ____________________

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CASE 14 Barsuk SBML — Simulation-Based Mastery Learning Dissemination from Northwestern to the VA

2009 – 2020s

1. Barsuk, Cohen, Feinglass, McGaghie, & Wayne (2009), “Use of simulation-based education to reduce catheter-related bloodstream infections,” Archives of Internal Medicine 169(15):1420–1423, doi:10.1001/archinternmed.2009.215. — Retrieved from: ____________________

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CASE 15 I-PASS Handoff Bundle — Structuring the Human-to-Human Transfer

2014

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